

Supporting Information

Supporting Table 1. All non-native fish species detected by eDNA, including taxonomy, origin, IUCN status, and the number of MPAs where each species was detected.

Species	Family	Origin	IUCN Status	Number of MPAs
<i>Alepes djedaba</i>	Carangidae	RS	LC	1
<i>Atherinomorus forskalii</i>	Atherinidae	RS	NA	4
<i>Callionymus filamentosus</i>	Callionymidae	RS	NA	3
<i>Cheilodipterus novemstriatus</i>	Apogonidae	RS	NA	1
<i>Ctenogobius boleosoma</i>	Gobiidae	A	LC	1
<i>Enchelycore anatina</i>	Muraenidae	A	LC	4
<i>Encrasicholina punctifer</i>	Engraulidae	RS	LC	1
<i>Equulites klunzingeri</i>	Leiognathidae	RS	NA	1
<i>Fistularia commersonii</i>	Fistulariidae	RS	LC	5
<i>Fistularia petimba</i>	Fistulariidae	A	LC	1
<i>Gymnammodytes semisquamatus</i>	Ammodytidae	A	LC	3
<i>Hemiramphus far</i>	Hemiramphidae	RS	NA	2
<i>Herklotsichthys punctatus</i>	Clupeidae	RS	LC	1
<i>Himantura uarnak</i>	Dasyatidae	RS	EN	1
<i>Jaydia smithi</i>	Apogonidae	RS	LC	1
<i>Lagocephalus scleratus</i>	Tetraodontidae	RS	LC	5
<i>Lagocephalus suezensis</i>	Tetraodontidae	RS	LC	2
<i>Nemipterus randalli</i>	Nemipteridae	RS	LC	1
<i>Ostorhinchus fasciatus</i>	Apogonidae	RS	LC	1
<i>Parexocoetus mento</i>	Exocoetidae	RS	NA	2
<i>Pterois miles</i>	Scorpaenidae	RS	LC	5
<i>Saurida lessepsianus</i>	Synodontidae	RS	LC	1
<i>Scomberomorus commerson</i>	Scombridae	RS	NT	2
<i>Siganus luridus</i>	Siganidae	RS	LC	3
<i>Solea senegalensis</i>	Soleidae	A	DD	2
<i>Synodus synodus</i>	Synodontidae	A	LC	1
<i>Torquigener flavimaculosus</i>	Tetraodontidae	RS	LC	4
<i>Upeneus moluccensis</i>	Mullidae	RS	LC	2
<i>Upeneus pori</i>	Mullidae	RS	LC	5

Supporting Table 2. Beta diversities between MPAs for Jaccard matrix used for the PCoAs. Not included are MPAs with zero non-native species; for these mean dissimilarity = 1, mean turnover = NA, mean nestedness = NA.

MPA 1	MPA 2	mean dissimilarity	mean turnover	mean nestedness
"20001"	"10001"	0.636363636363636	0.428571428571429	0.207792207792208
"20002"	"10001"	0.826086956521739	0.666666666666667	0.159420289855072
"20003"	"10001"	0.875	0.769230769230769	0.105769230769231
"30001"	"10001"	0.96	0.923076923076923	0.0369230769230769
"30002"	"10001"	1	1	0
"40001"	"10001"	0.95	0.666666666666667	0.283333333333333
"40002"	"10001"	0.95	0.666666666666667	0.283333333333333
"70001"	"10001"	1	1	0
"70004"	"10001"	1	1	0
"70005"	"10001"	1	1	0
"90004"	"10001"	1	1	0
"90006"	"10001"	1	1	0
"90008"	"10001"	1	1	0
"20002"	"20001"	0.416666666666667	0.222222222222222	0.194444444444444
"20003"	"20001"	0.642857142857143	0.545454545454545	0.0974025974025974
"30001"	"20001"	0.714285714285714	0.6	0.114285714285714
"30002"	"20001"	0.916666666666667	0.666666666666667	0.25
"40001"	"20001"	0.818181818181818	0	0.818181818181818
"40002"	"20001"	0.916666666666667	0.666666666666667	0.25
"70001"	"20001"	1	1	0
"70004"	"20001"	1	1	0
"70005"	"20001"	1	1	0
"90004"	"20001"	1	1	0
"90006"	"20001"	1	1	0
"90008"	"20001"	1	1	0
"20003"	"20002"	0.545454545454545	0.545454545454545	0
"30001"	"20002"	0.636363636363636	0.6	0.0363636363636364
"30002"	"20002"	0.888888888888889	0.666666666666667	0.222222222222222
"40001"	"20002"	0.75	0	0.75
"40002"	"20002"	0.888888888888889	0.666666666666667	0.222222222222222
"70001"	"20002"	1	1	0
"70004"	"20002"	1	1	0
"70005"	"20002"	1	1	0
"90004"	"20002"	1	1	0
"90006"	"20002"	1	1	0
"90008"	"20002"	1	1	0
"30001"	"20003"	0.333333333333333	0.25	0.0833333333333333
"30002"	"20003"	0.75	0	0.75
"40001"	"20003"	0.888888888888889	0.666666666666667	0.222222222222222
"40002"	"20003"	0.888888888888889	0.666666666666667	0.222222222222222
"70001"	"20003"	1	1	0
"70004"	"20003"	0.875	0	0.875
"70005"	"20003"	1	1	0
"90004"	"20003"	1	1	0
"90006"	"20003"	1	1	0
"90008"	"20003"	1	1	0
"30002"	"30001"	0.714285714285714	0	0.714285714285714
"40001"	"30001"	0.875	0.666666666666667	0.208333333333333
"40002"	"30001"	0.875	0.666666666666667	0.208333333333333
"70001"	"30001"	1	1	0
"70004"	"30001"	0.857142857142857	0	0.857142857142857
"70005"	"30001"	1	1	0
"90004"	"30001"	1	1	0
"90006"	"30001"	1	1	0

Supporting Table 2. – continued from previous page

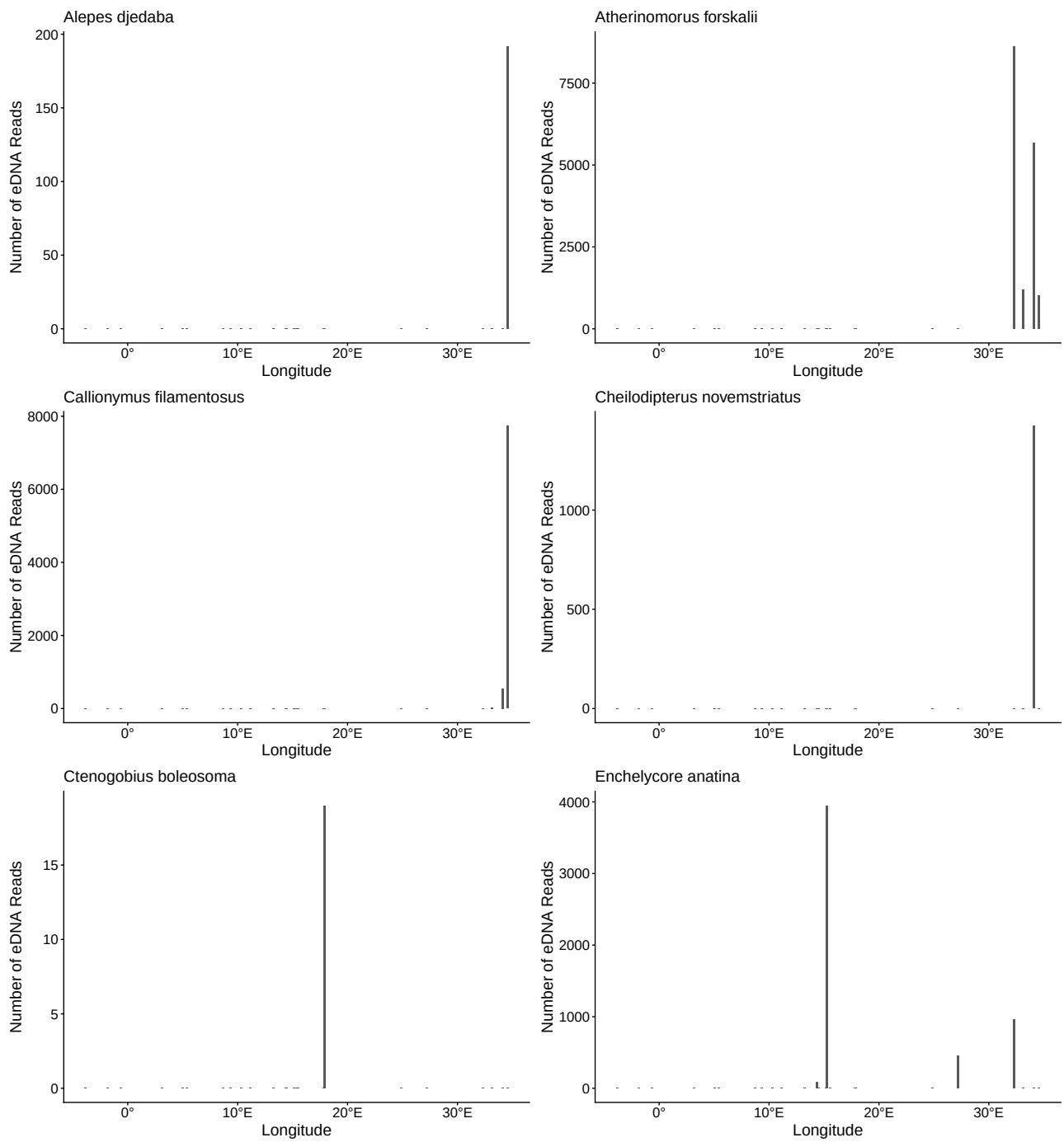
MPA 1	MPA 2	mean dissimilarity	mean turnover	mean nestedness
"90008"	"30001"	1	1	0
"40001"	"30002"	1	1	0
"40002"	"30002"	1	1	0
"70001"	"30002"	1	1	0
"70004"	"30002"	1	1	0
"70005"	"30002"	1	1	0
"90004"	"30002"	1	1	0
"90006"	"30002"	1	1	0
"90008"	"30002"	1	1	0
"40002"	"40001"	0.666666666666667	0.666666666666667	0
"70001"	"40001"	1	1	0
"70004"	"40001"	1	1	0
"70005"	"40001"	1	1	0
"90004"	"40001"	1	1	0
"90006"	"40001"	1	1	0
"90008"	"40001"	1	1	0
"70001"	"40002"	1	1	0
"70004"	"40002"	0.5	0	0.5
"70005"	"40002"	1	1	0
"90004"	"40002"	1	1	0
"90006"	"40002"	1	1	0
"90008"	"40002"	1	1	0
"70004"	"70001"	1	1	0
"70005"	"70001"	1	1	0
"90004"	"70001"	1	1	0
"90006"	"70001"	1	1	0
"90008"	"70001"	1	1	0
"70005"	"70004"	1	1	0
"90004"	"70004"	1	1	0
"90006"	"70004"	1	1	0
"90008"	"70004"	1	1	0
"90004"	"70005"	1	1	0
"90006"	"70005"	0	0	0
"90008"	"70005"	0.5	0	0.5
"90006"	"90004"	1	1	0
"90008"	"90004"	0.5	0	0.5
"90008"	"90006"	0.5	0	0.5

Supporting Table 3. Results of the ORMEF database GLMM for each predictor. Estimates, standard errors, test statistics, p-values, and Lower and upper 95% confidence limits are shown for each predictor.

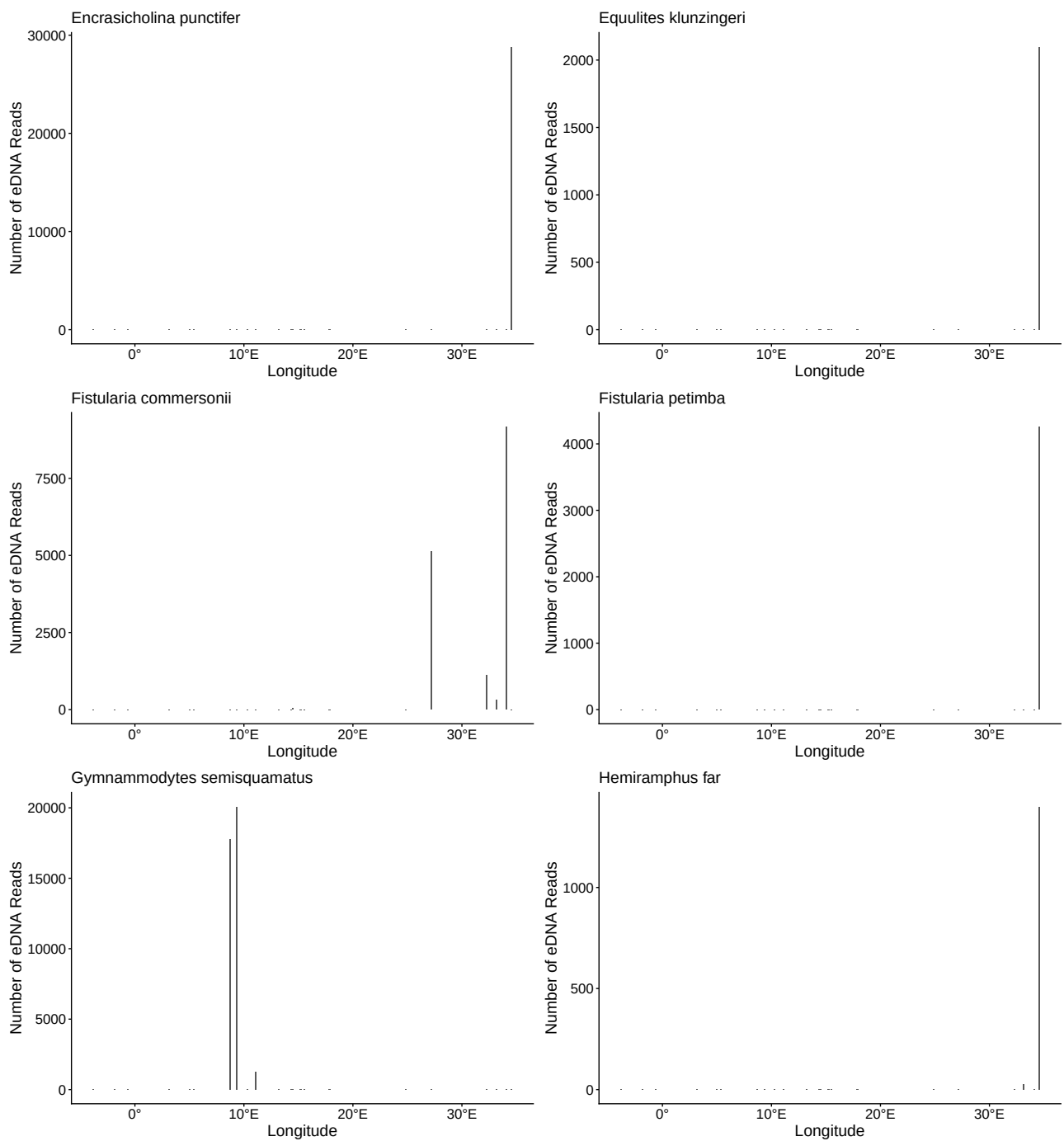
predictor	estimate	std.error	statistic	p.value	AIC	conf.low	conf.high
Mean Sea Surface Temperature (SST)	0.31630337	0.025031767	12.6360784	1.33550493596404e-36	5165.393811	0.267241107	0.365365634
Mean Sea Surface Salinity (SSS)	0.108043	0.021964865	4.918901182	8.70313898325815e-07	5295.916451	0.064991865	0.151094135
Human Gravity	0.019923659	0.023971391	0.831143229	0.405892721	5217.631918	-0.027060267	0.066907586
Fishing Gravity	-0.053303737	0.022725841	-2.345512126	0.019000961	5318.596531	-0.097846386	-0.008761087
Circulation Cost-Distance to Red Sea	-0.308826136	0.020905794	-14.77227498	2.21121210413764e-49	5111.710512	-0.349801493	-0.26785078

Supporting Table 4. Results of the eDNA GLMM for each predictor. Estimates, standard errors, test statistics, p-values, and Lower and upper 95% confidence limits are shown for each predictor.

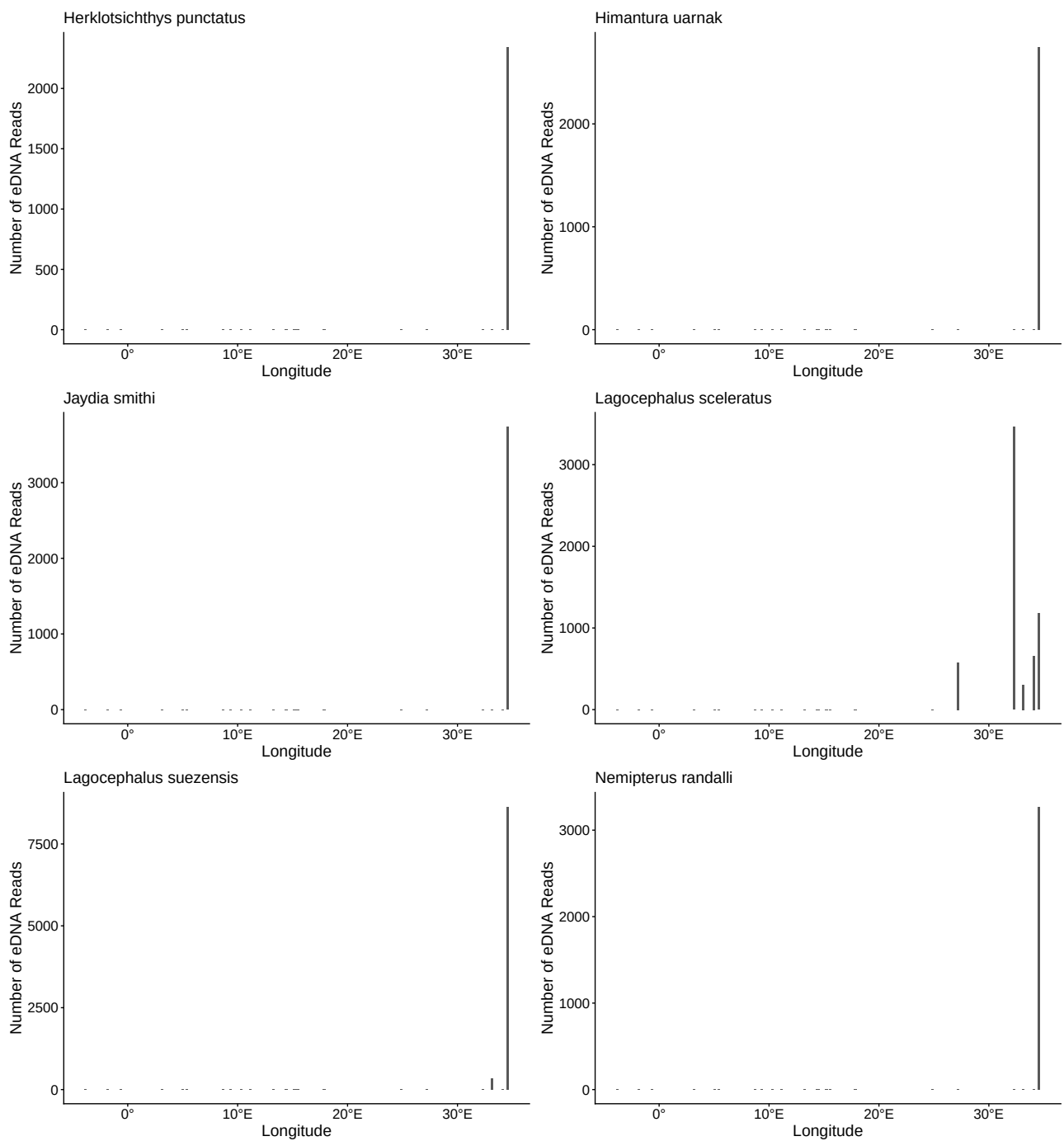
predictor	estimate	std.error	statistic	p.value	AIC	conf.low	conf.high
Mean Sea Surface Temperature (SST)	1.564155885	0.191432028	8.170816047	3.06310387734567e-16	200.5293432	1.18894911	1.93936266
Mean Sea Surface Salinity (SSS)	1.194094976	0.375802581	3.177452831	0.001485749	218.803876	0.457521917	1.930668036
Human Gravity	-0.181733316	0.341580909	-0.532035929	0.594701111	228.0833233	-0.851231899	0.487765266
Fishing Gravity	-0.224960834	0.474988461	-0.473613261	0.635775717	228.180717	-1.155938219	0.70601655
Circulation Cost-Distance to Red Sea	-1.618760216	0.135660907	-11.93240008	8.0226549194793e-33	187.0193008	-1.884655594	-1.352864839



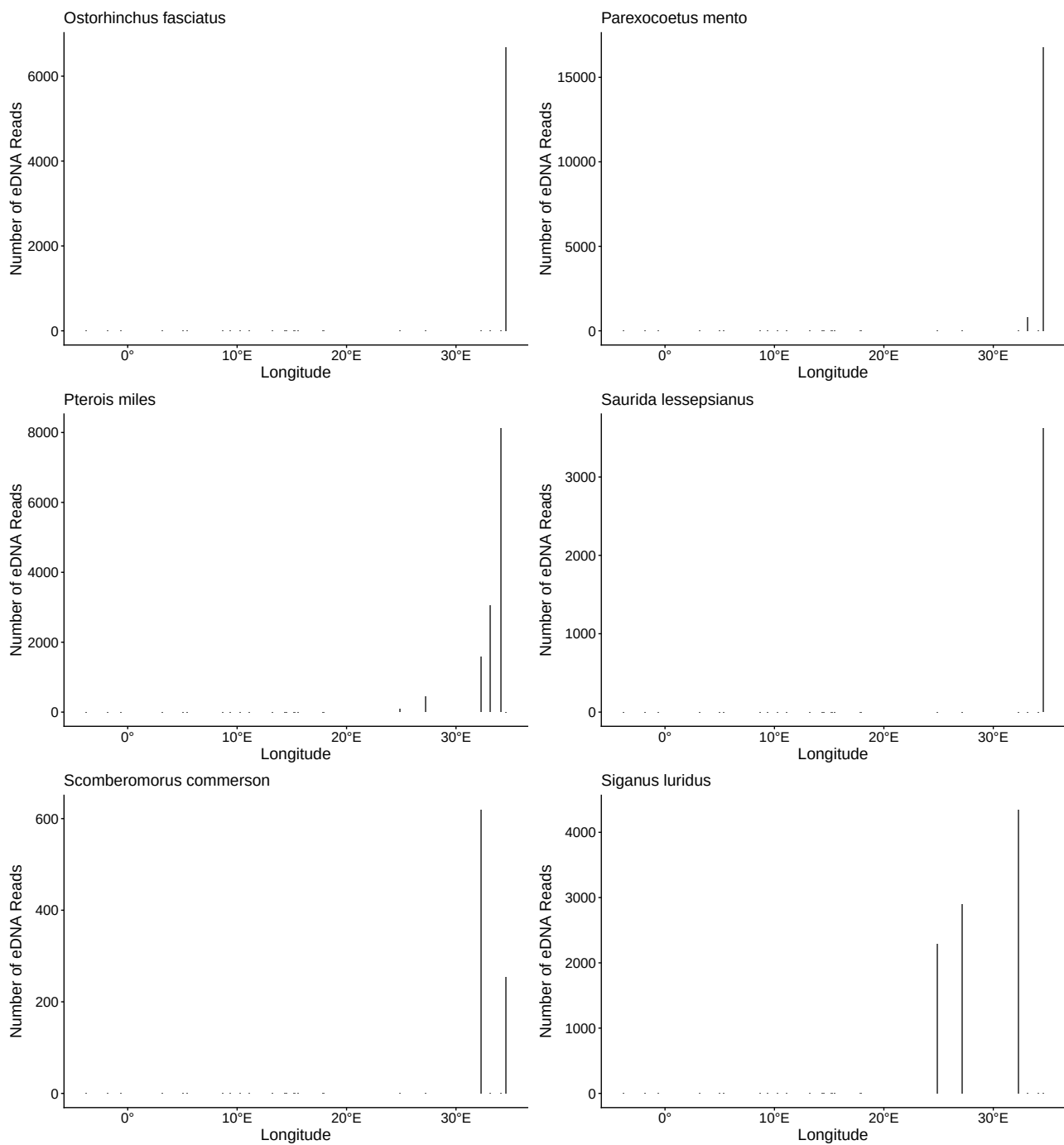
Supporting Figure 1. eDNA read counts for each non-native species along the longitude: Page 1



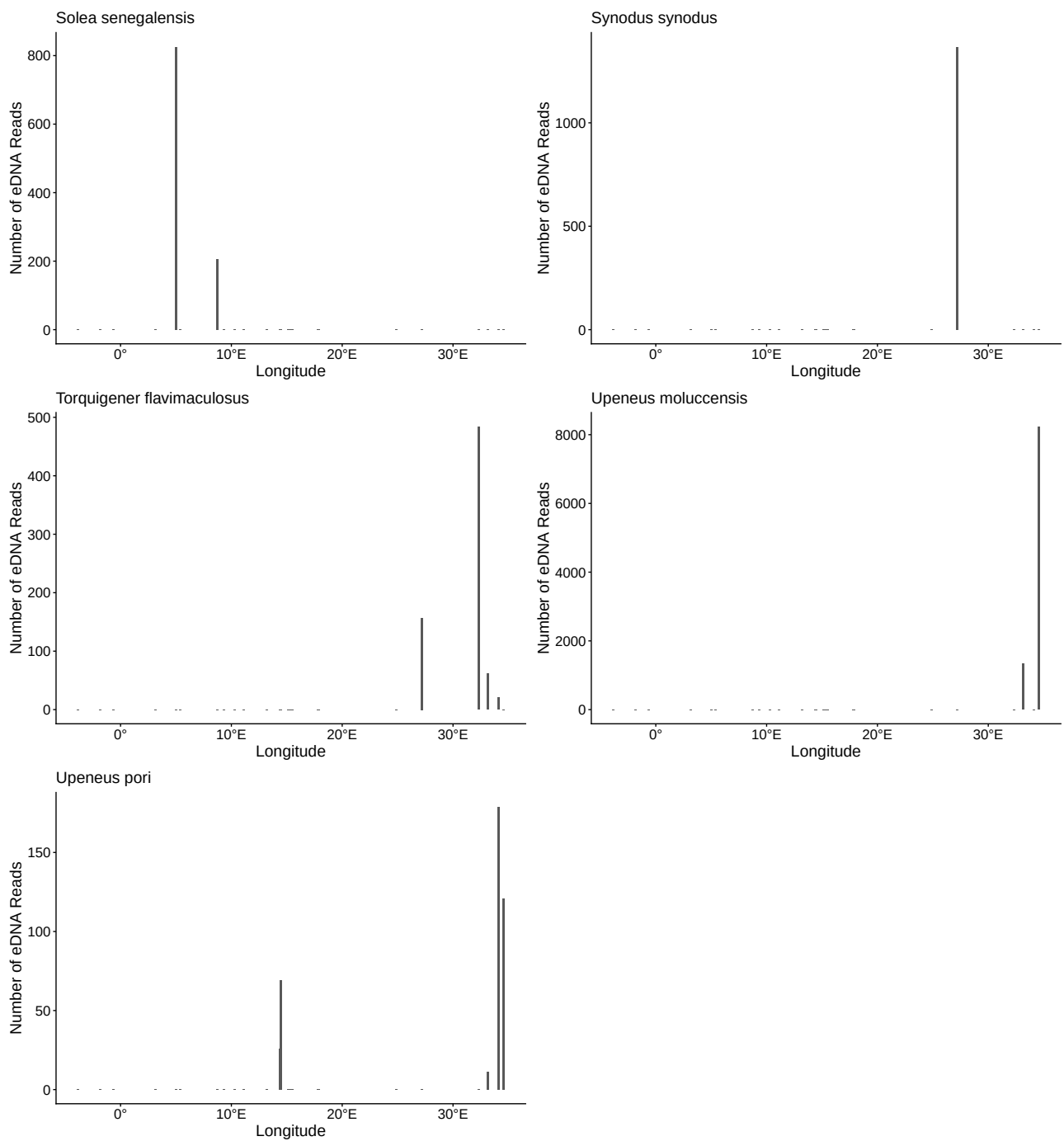
Supporting Figure 2. Supporting Figure 1 – continued: Page 2



Supporting Figure 3. Supporting Figure 1 – continued: Page 3

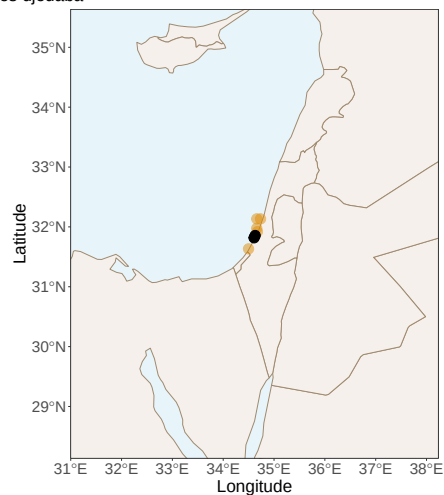


Supporting Figure 4. Supporting Figure 1 – continued: Page 4

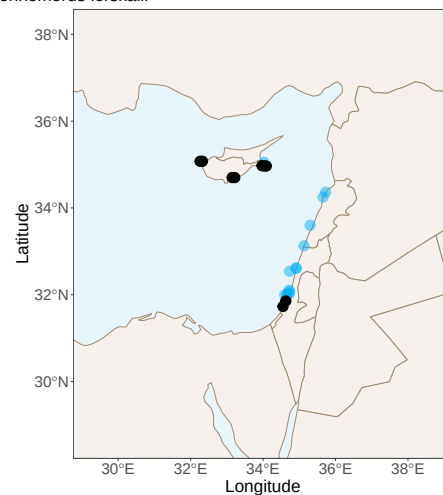


Supporting Figure 5. Supporting Figure 1 – continued: Page 5

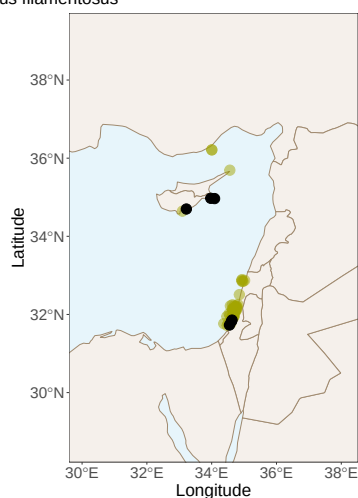
Alepes djedaba



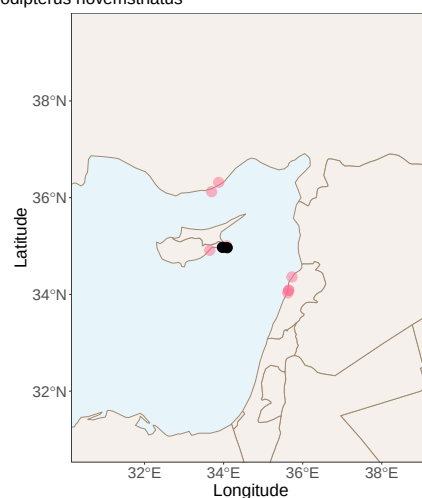
Atherinomorus forskalii



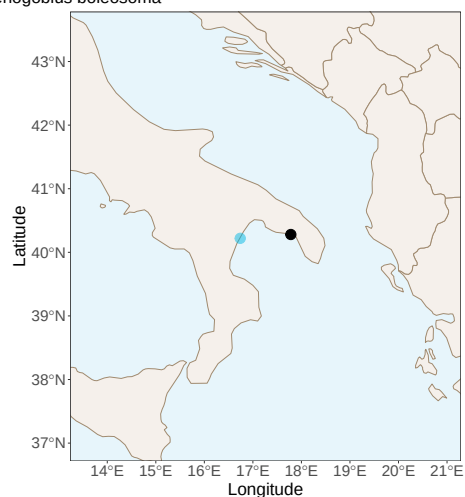
Callionymus filamentosus



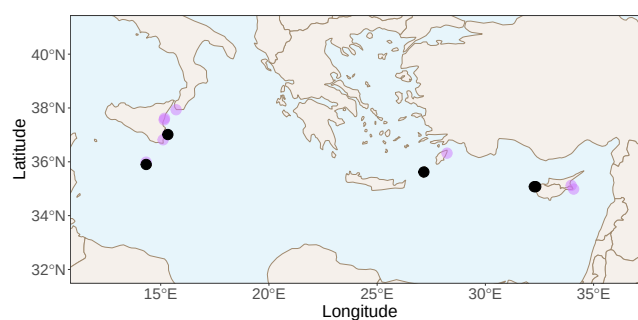
Cheilodipterus novemstriatus



Ctenogobius boleosoma

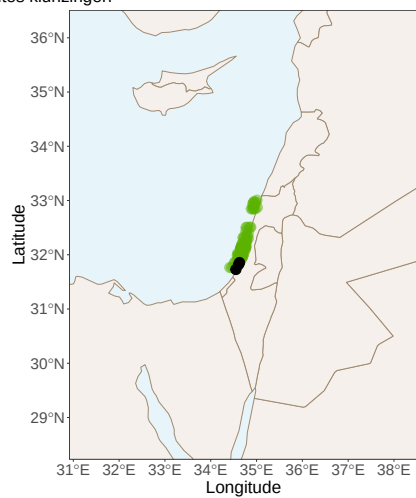


Enchelycore anatina

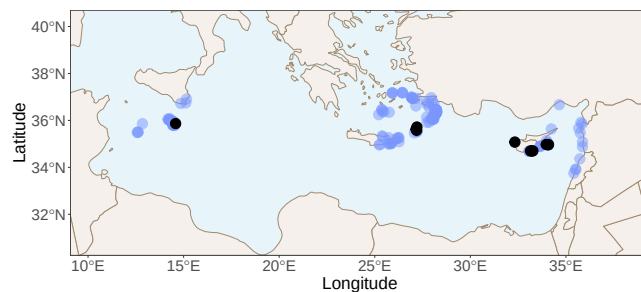


Supporting Figure 6. Maps of the Mediterranean Sea for each non-native species, showing in black the collected samples and, within a 200 km radius, in colour, the observations from the ORMEF database: Page 1

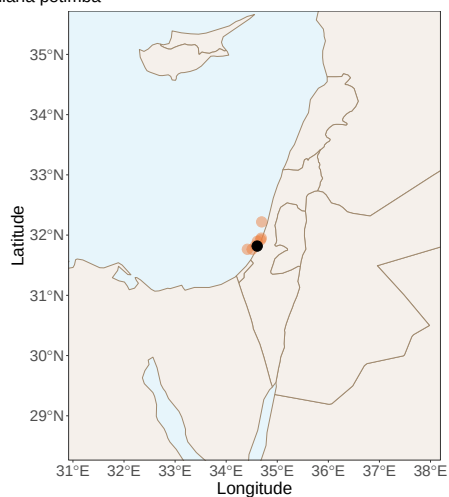
Equulites klunzingeri



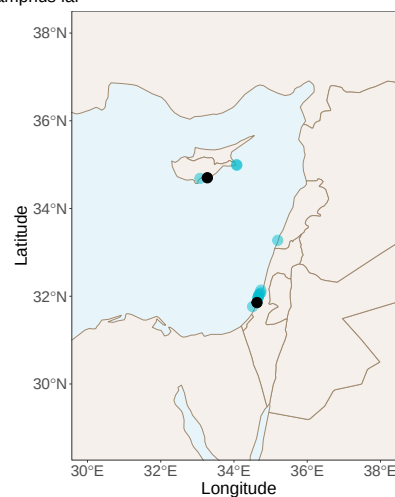
Fistularia commersonii



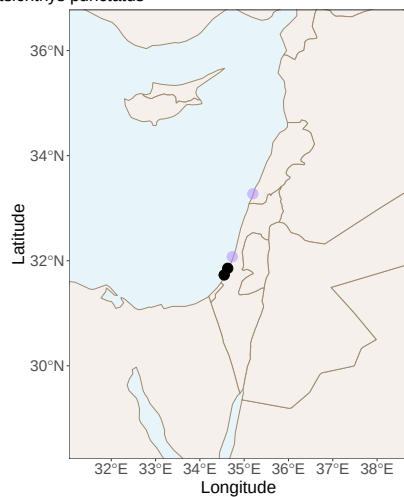
Fistularia petimba



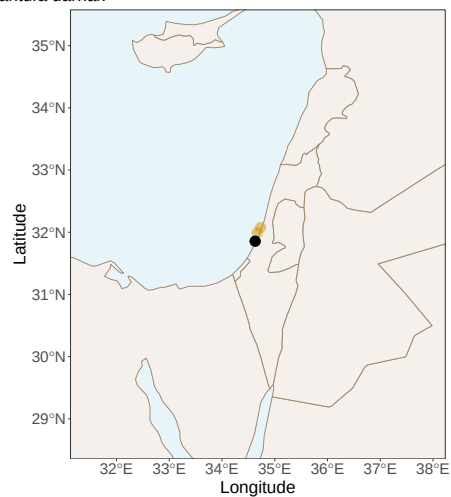
Hemiramphus far



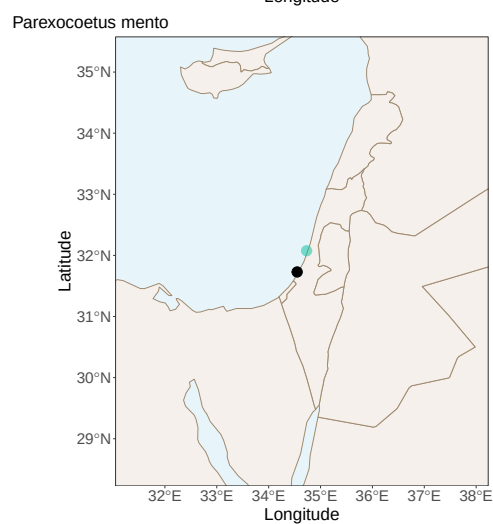
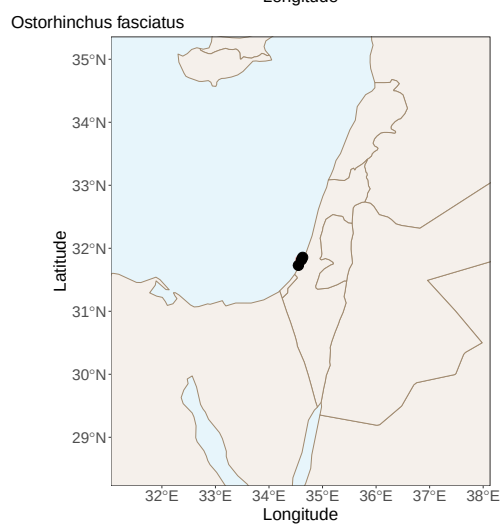
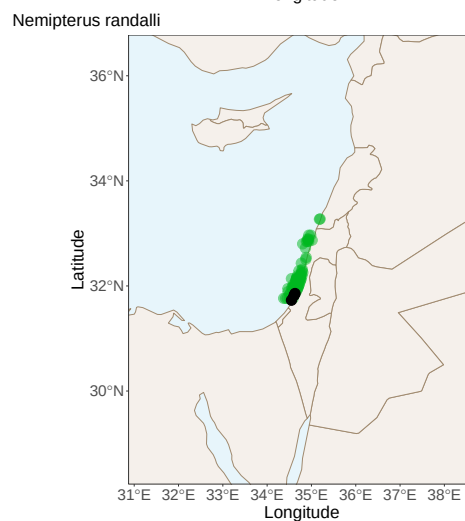
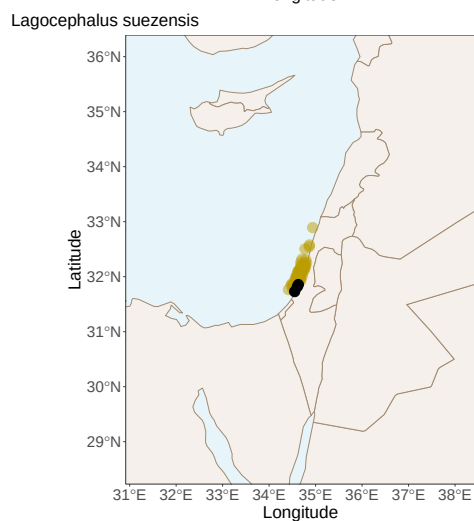
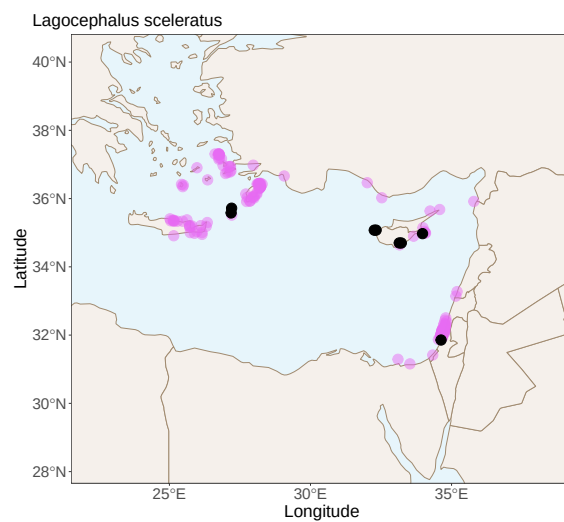
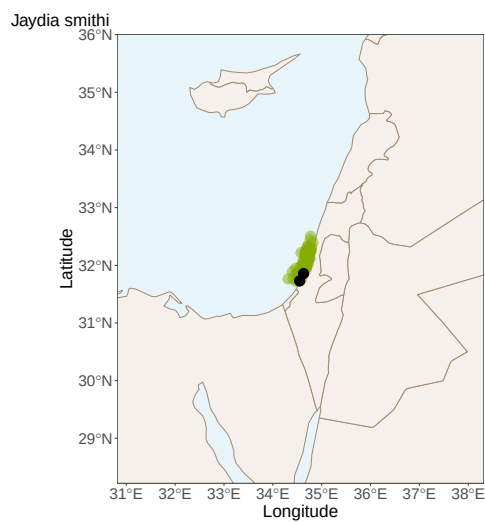
Herklotsichthys punctatus



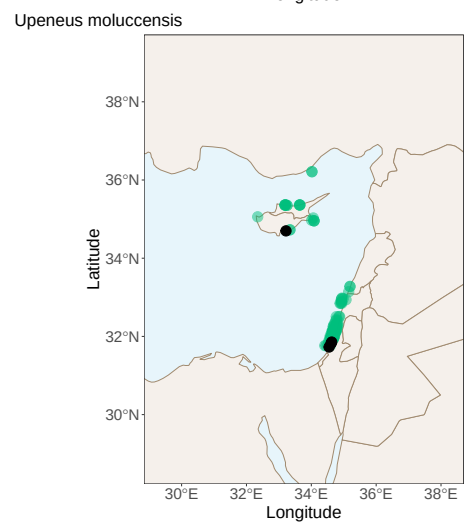
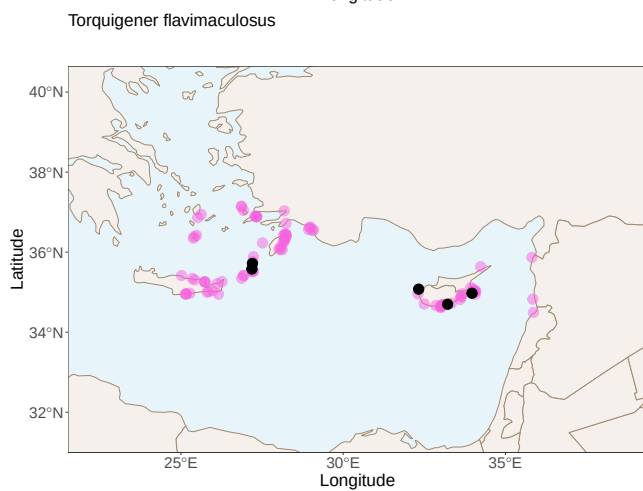
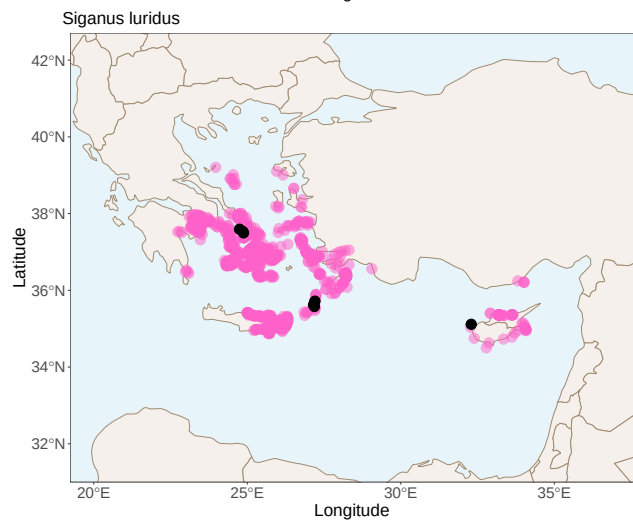
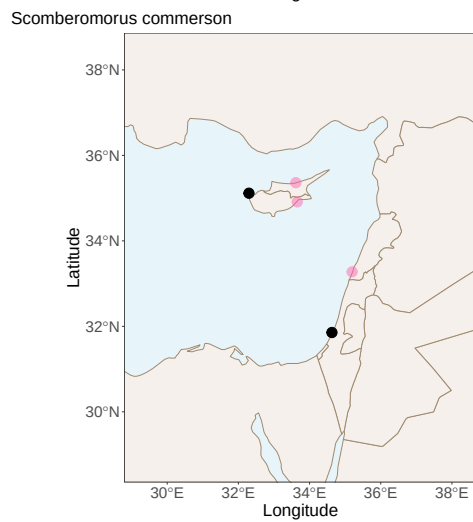
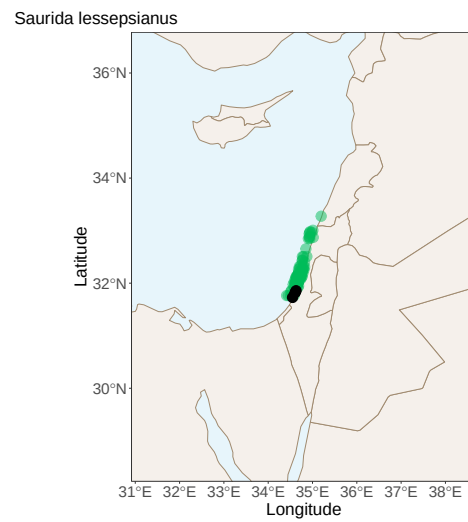
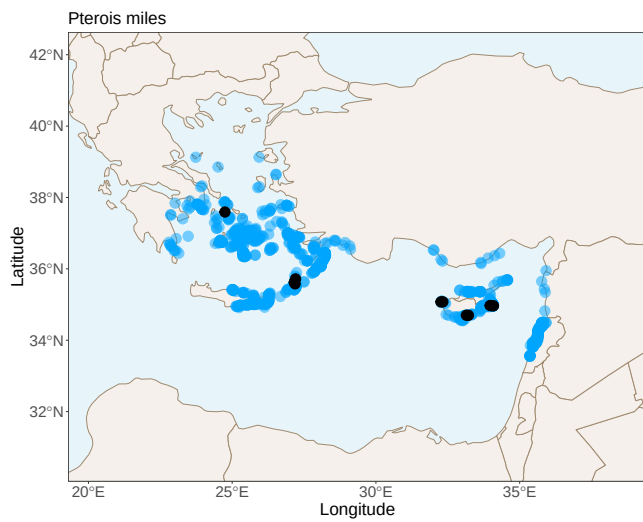
Himantura uarnak



Supporting Figure 7. Supporting Figure 6 – continued: Page 2

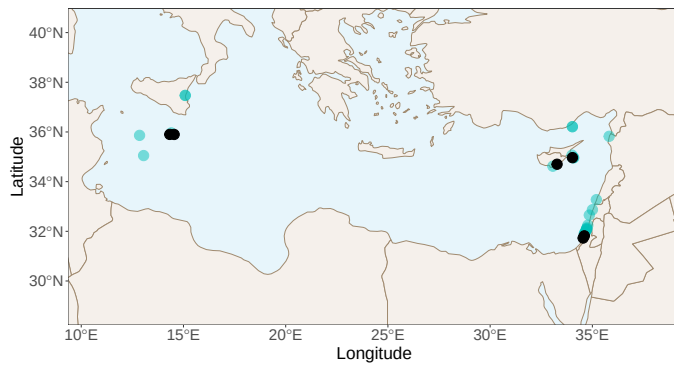


Supporting Figure 8. Supporting Figure 6 – continued: Page 3



Supporting Figure 9. Supporting Figure 6 – continued: Page 4

Upeneus pori



Supporting Figure 10. Supporting Figure 6 – continued: Page 5